

MYSQL DATABASE MANAGEMENT SYSTEM

LAB SHEET #2

Title:

- Database constraints and relational integrity using primary key, foreign key and constraints

Objectives:

- To be familiar with primary key, foreign key and other keys
- To be familiar with database constraints

Background Concept:

- Primary Key:** A column that uniquely identifies each row in a table. It cannot contain NULL values and must be unique.
 - ALTER TABLE <table_name> ADD PRIMARY KEY (<column_name>);
- Foreign Key:** A column in one table that refers to the primary key in another table, creating a link between the two.
 - ALTER TABLE <table_name> ADD CONSTRAINT <constraint_name> FOREIGN KEY (<column_name>) REFERENCES <table_name>(<column_name>) ON DELETE CASCADE ON UPDATE CASCADE;
- Unique Key:** Enforces that all values in a column are unique, but unlike a primary key, it can contain a single NULL value.
 - ALTER TABLE <table_name> ADD CONSTRAINT <constraint_name> UNIQUE (<column_name>, <column_name>);
- Candidate Key:** A minimal set of attributes that can uniquely identify a row. A table can have multiple candidate keys, and one is chosen as the primary key.
- Super Key:** A set of one or more attributes that uniquely identifies all the records in a table. A candidate key is a type of super key that has no redundant attributes.
- Composite Key:** A key that consists of two or more columns to uniquely identify a record.

Lab works:

- Create database named "EMPLOYEE_MGMT"
- Create EMPLOYEE and DEPARTMENT tables with following relational database models with following constraints
 - Make empno and did primary key
 - Make departid foreign key that references did from department table
 - Make licensno and departcode unique and not null

empno	empname	designation	departid	licensno	gender	salary
5004	Sameer Sharma	Officer	301	323-525	m	38000
5010	Rakshya Acharya	Manager	303	987-321	f	45000
5005	Krishna Devkota	Officer	301	547-685	m	38000
5009	Ram Yadav	Salesperson	304	342-154	m	35000
5003	Sunita Chaudhary	Salesperson	305	241-654	f	35000

did	departname	departcode	email
301	Information Technology	IT	it@ourmail.com
302	Human Resource	HR	hr@ourmail.com
303	Finance	FI	fi@ourmail.com
304	Marketing	MK	mk@ourmail.com
305	Sales	SL	sl@ourmail.com

3. Add constraint that ensures salary is always greater than 15000
4. Add constraint to set default value 302 for departid
5. Add details of Account department as follows

did	departname	departcode	email
306	Account	AC	ac@ourmail.com

6. Add new record of an employee working on account department
7. Change the department of empno 5010 to account department
8. Change the did to 308
9. Display empno, empname, departid, salary from employee table
10. Delete account department and view the records of employee.
11. Create tables with following schema using proper constraints and rules
 - a. Salesman (s_id, name, city, email)
 - b. Orders (ord_no, prch_amt, ord_date, c_id, s_id)
 - c. Customer (c_id, name, city, contact)